Software Design Specifications

UNIVERSITY OF SARGODHA

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF COMPUTING & IT

Bachelor's Degree in Computer Science

Area: AI & ML

Project Code: HiGWM-22

Physics Informed LLM Based Weather Prediction System

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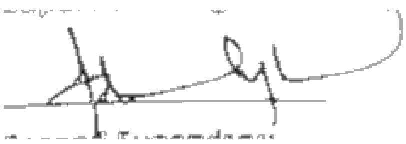

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Definition of Terms, Acronyms and Abbreviations

This section provides the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
NWP	Numerical Weather Prediction
PINNS	Physics Informed Neural Network System
LLMs	Large Language Model
ML	Machine Learning

1. Introduction

1.1. Purpose of Document

The purpose is of this document to present the design requirements of this project "Physics Informed LLM Based Weather Prediction System". It will provide a detailed explanation of system's component design, data flow, architecture and implementation strategies. To ensure an alignment between project requirements and design decisions for stakeholders, this document will assist as a reference and technical blueprint for developers. The intendent audience (project advisors, the project manager, the development team members, and the university evaluators) who will evaluate the system's planed architecture and feasibility.

1.2. Project Overview

The main concern of this project is to create a hybrid weather prediction system that integrates Physics-Informed Neural Network (PINNS) with Machine Learning (ML) techniques and Large Language Models (LLMs). Traditional Numerical Weather Prediction (NWP) models are accurate and precise but require major computational power and time which may often lead to delays in real-time forecasting. On the other-hand, pure data driven ML models can deliver faster results but may generate outputs that breach physical laws for an instant unrealistic temperature point or wind pattern. In contrast, our system will aim to address these issues by integrating physical equations (e.g., Navier-Stokes for fluid dynamics) directly into the neural network training process through PINNs, allowing predictions are both data informed and physically consistent. LLMs will then process these outputs to produce natural language summaries, allowing the forecasts accessible to non experts. The main concern is to improve accuracy for short term predictions, less computational cost a compared to supercomputer based NWP and enhanced interpretability which can help in applications such as agriculture, disaster management, and policy planning.

1.3. Scope

The system will handle short-term weather forecasting (1-3 days ahead) for key variables like temperature, humidity, pressure, and wind speed, using publicly available datasets. It includes data preprocessing, PINN-based prediction with physics constraints, LLM integration for human-readable outputs, and basic evaluation against ML baselines. The system will be a prototype focused on demonstration, not production deployment. What the system will do: Process weather data, generate physics-consistent forecasts, and provide interpretable summaries. What the system will not do: Perform long-term climate modeling, high-resolution global simulations, real-time data ingestion from sensors, or serve as a public-facing operational tool. Exclusions also include training custom LLMs from scratch or using proprietary datasets.

The system will generate short term weather forecasting (1-3 days ahead) for key variables like temperature, pressure, humidity and wind speed using publicly available datasets.

2. Design Considerations

2.1. Assumptions

- Datasets like ERA5 are freely accessible.
- Minimum hardware requirements for Python/ML libraries.
- Pre-trained LLMs offer reliable explanations.

2.2. Dependencies

- External Libraries e.g., PyTorch, Hugging Face Transformers etc.
- Availability of team resource.

2.3. Risks and Volatile Areas

2.3.1. Data Quality and Availability Risks

Risk: There is a risk of dataset going temporarily unavailable and may have missing values.

Design Response:

- Implement robust data validation and cleaning pipelines
- Include missing value imputation strategies
- Provide fallback to alternative datasets if primary source fails

2.3.2. Physics Constraint Implementation Complexity

Risk: Encoding physics law (Navier-Stokes, thermodynamic equations) into neural network is a very complex task.

Design Response:

- Start with simplified physics constraints and incrementally add complexity.
- Use established libraries like DeepXDE that provide physics-informed loss implementations.
- Implement modular physics constraint functions that can be individually tested.
- Allow configurable weighting between data loss and physics loss.

2.3.3. Model Performance and Accuracy

Risk: Achieving target accuracy can be a hypothetical thing or may perform worse than baseline ML models

Design Response:

- Implement comprehensive evaluation framework with multiple metrics.
- Design modular architecture allowing easy swapping of model components.
- Include extensive hyperparameter tuning capabilities.
- Compare against multiple baselines (LSTM, CNN, pure ML models).

2.3.4. Computational Resource Limitations

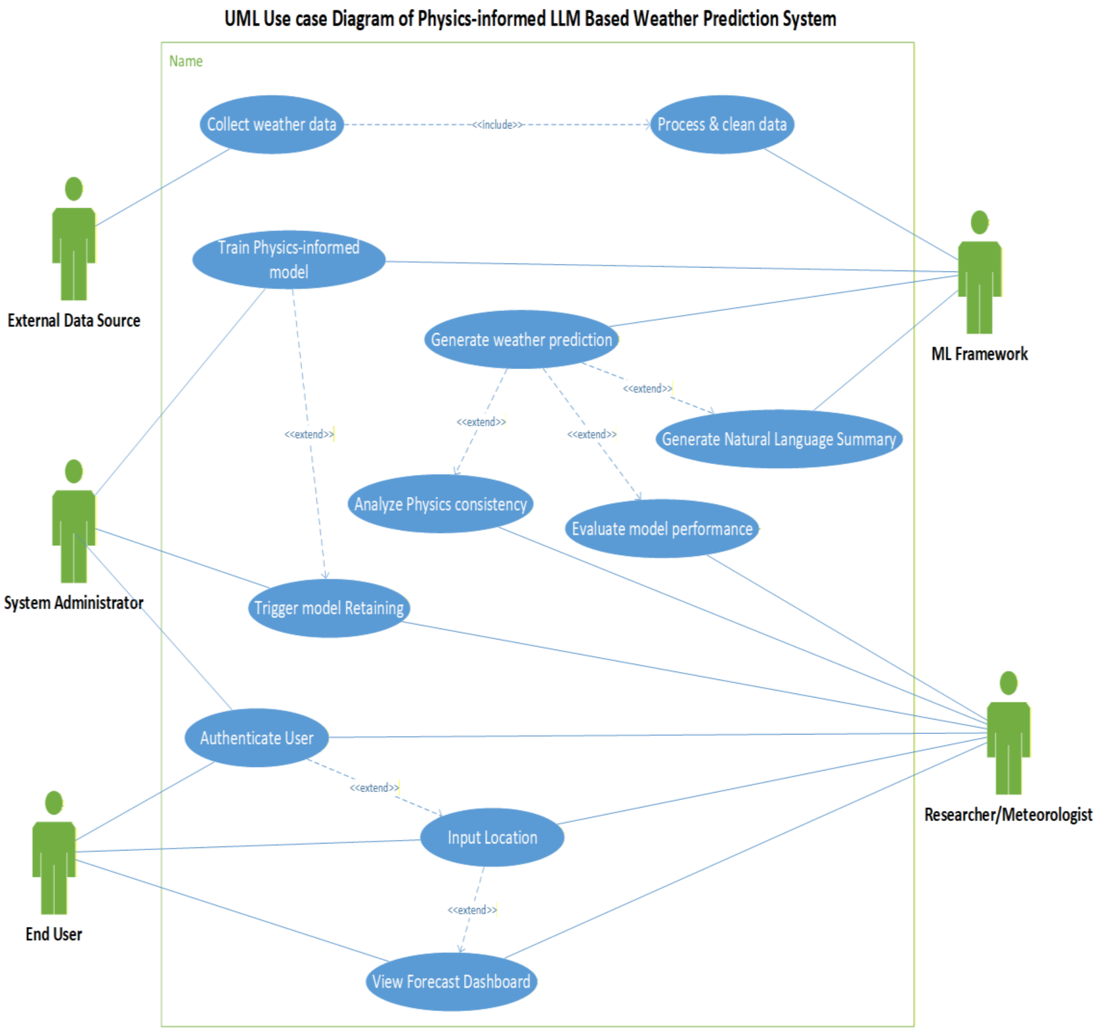
Risk: Training PINNs with physics constraints may require more computational resources than available on moderate hardware.

Design Response:

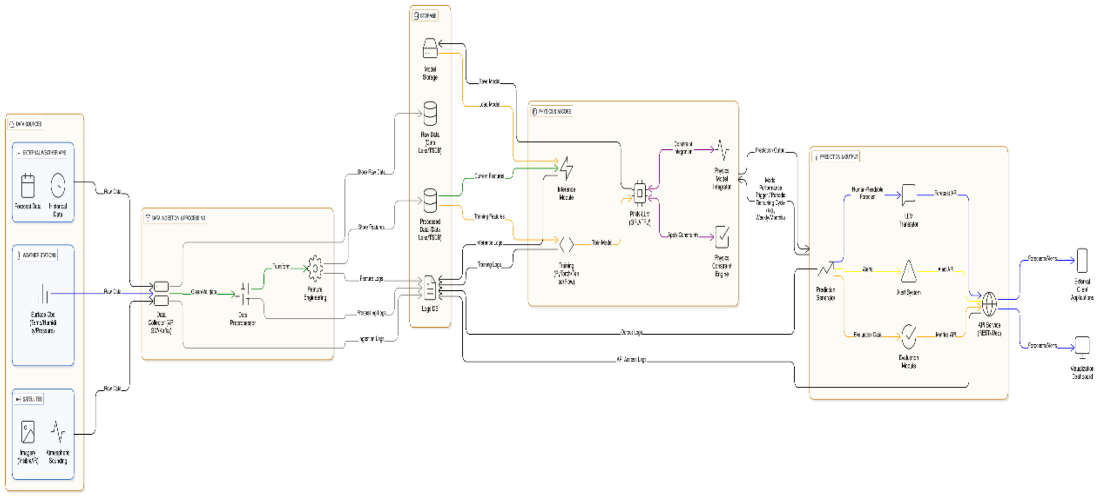
- Implement efficient data batching and sampling strategies
- Use mixed-precision training to reduce memory footprint
- Design scalable architecture that can run on CPU with degraded performance Provide checkpointing to resume training after interruptions
- Optimize grid resolution and temporal sampling

3. System Architecture

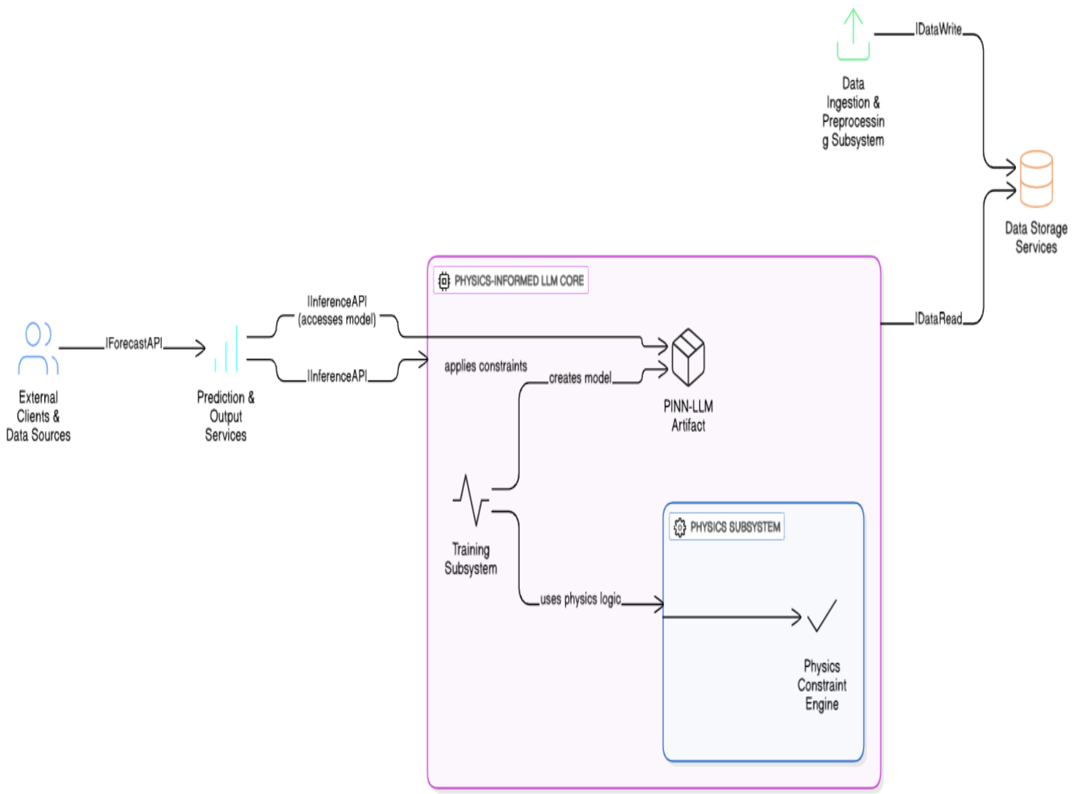
3.1. Use Case Diagram



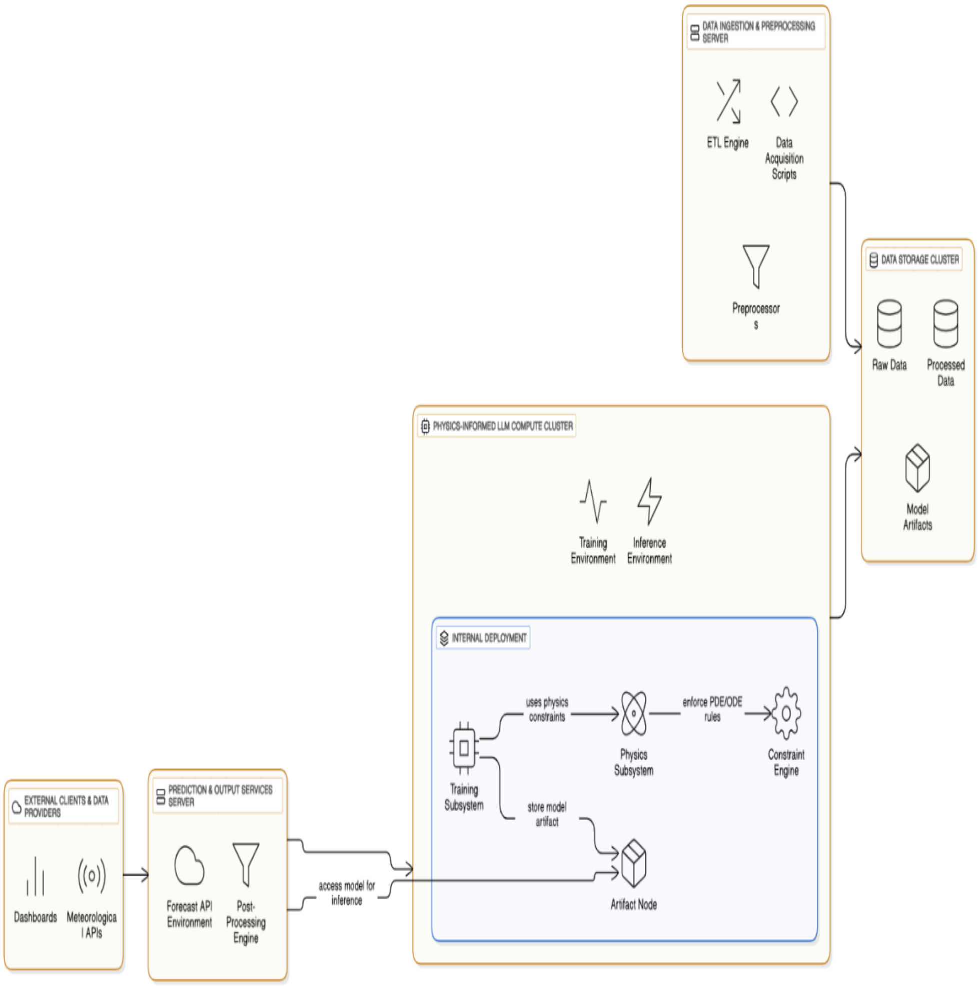
3.2. System Level Architecture



3.3. Component Diagram



3.4. Deployment Diagram



4. Design Strategies

4.1. Modular Layered Application Architecture

The proposed system will follow a **modular layered architecture**. This ensures that our system is highly scalable, maintainable, and to allow easy integration. The application is divided into three layers:

- **Presentation Layer:**
  It will provide an **application-based user interface** that will allow users to input forecast parameters such as location, date and time etc., and view numerical predictions and predictions in natural language.

- **Application Layer:**
  This step includes the core logic of the system, it will integrate the **Physics-Informed Neural Network (PINN)** for weather prediction and the **Large Language Model (LLM)** for generating human-readable forecasts. It will manage workflow execution and result processing.

- **Data Layer:**
  Preprocessed datasets, trained model parameters, and forecast outputs stored in structured formats such as **JSON or CSV** will be handeld.

This separation simplifies testing and supports future enhancements.

4.2. API-Driven Application Design

The application will use **Python APIs** will be used by the application for communication between different modules. These APIs handle model inference, data access, and delivery of forecast results, and will also be helpful in enabling easy integration with future application features.

4.3. Reusability and Extensibility

The system modules will be designed to be **independent and reusable**. In order to support additional weather variables, regions, the PINN and LLM components can be replaced or extended.

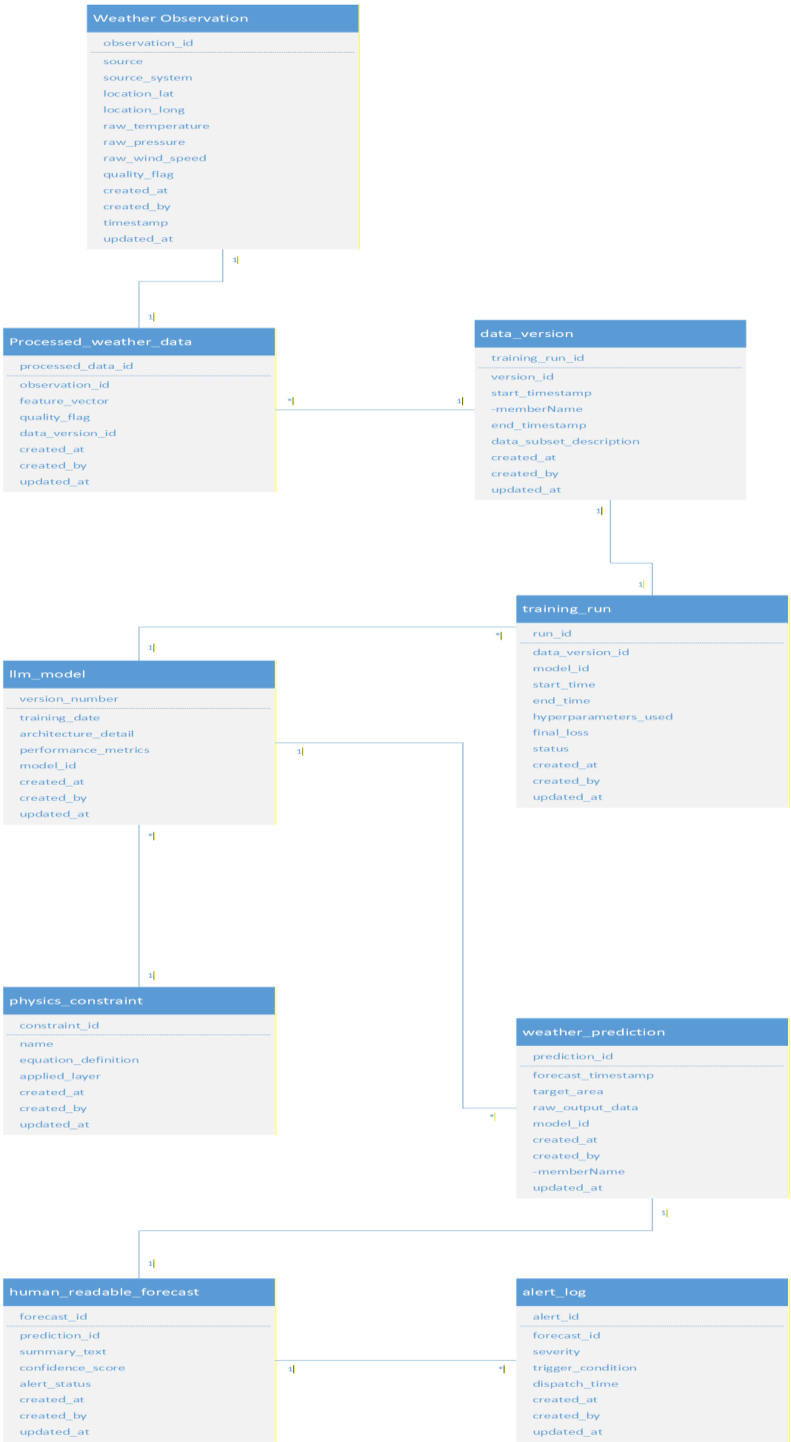
4.4. Security and Data Integrity

The application will access datasets through **secure connections** and data integrity will be ensured using validation checks. For ensuring privacy and safe academic usage, personal user data won’t be collected.

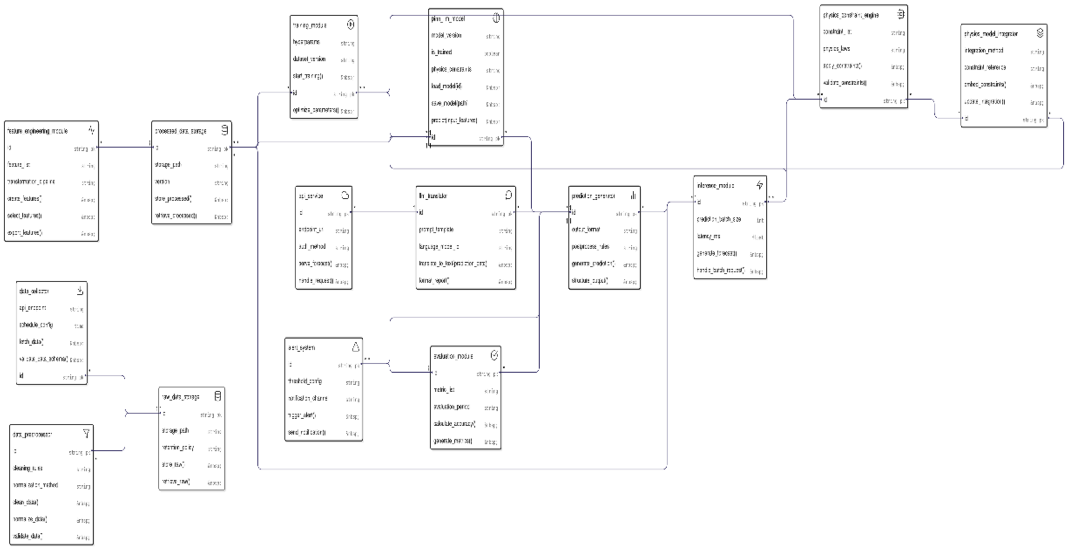
5. Detailed System Design

5.1. Class Diagram

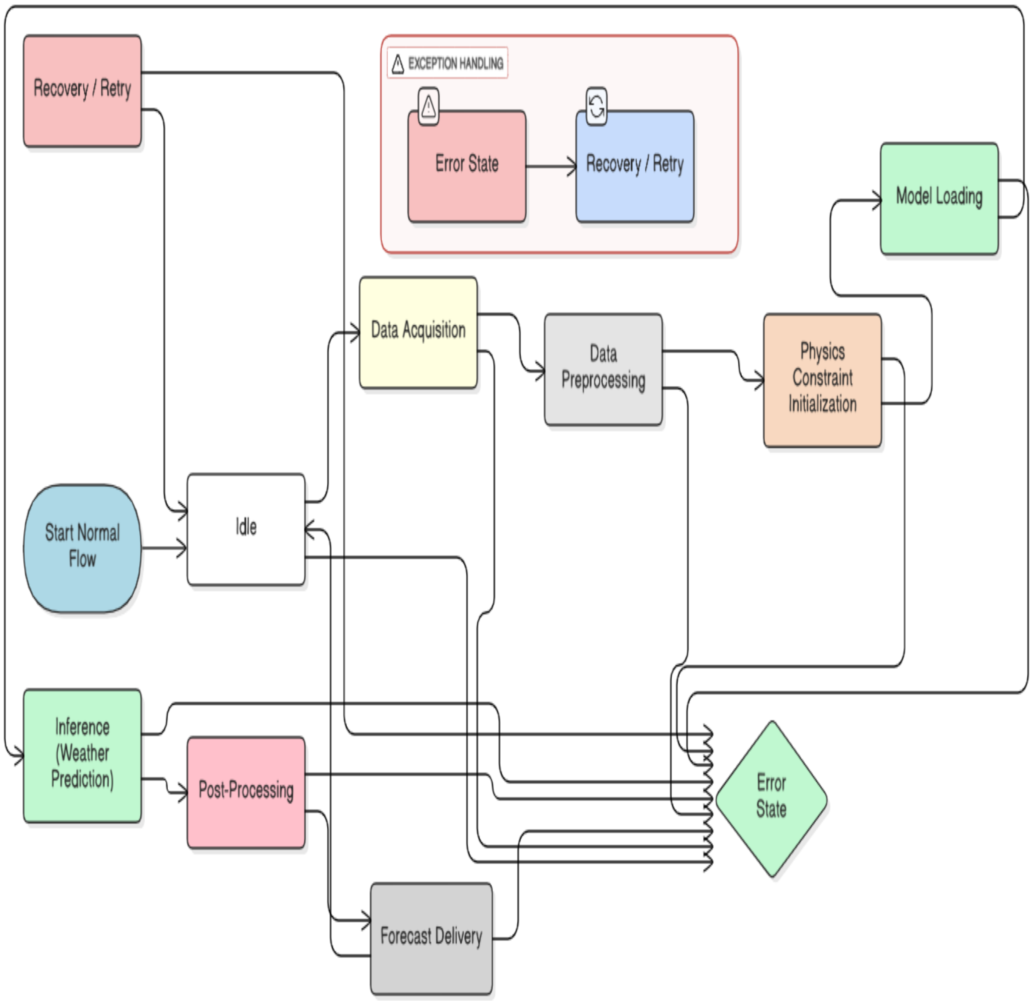
Class Diagram of Physics-informed LLM Based Weather Prediction System



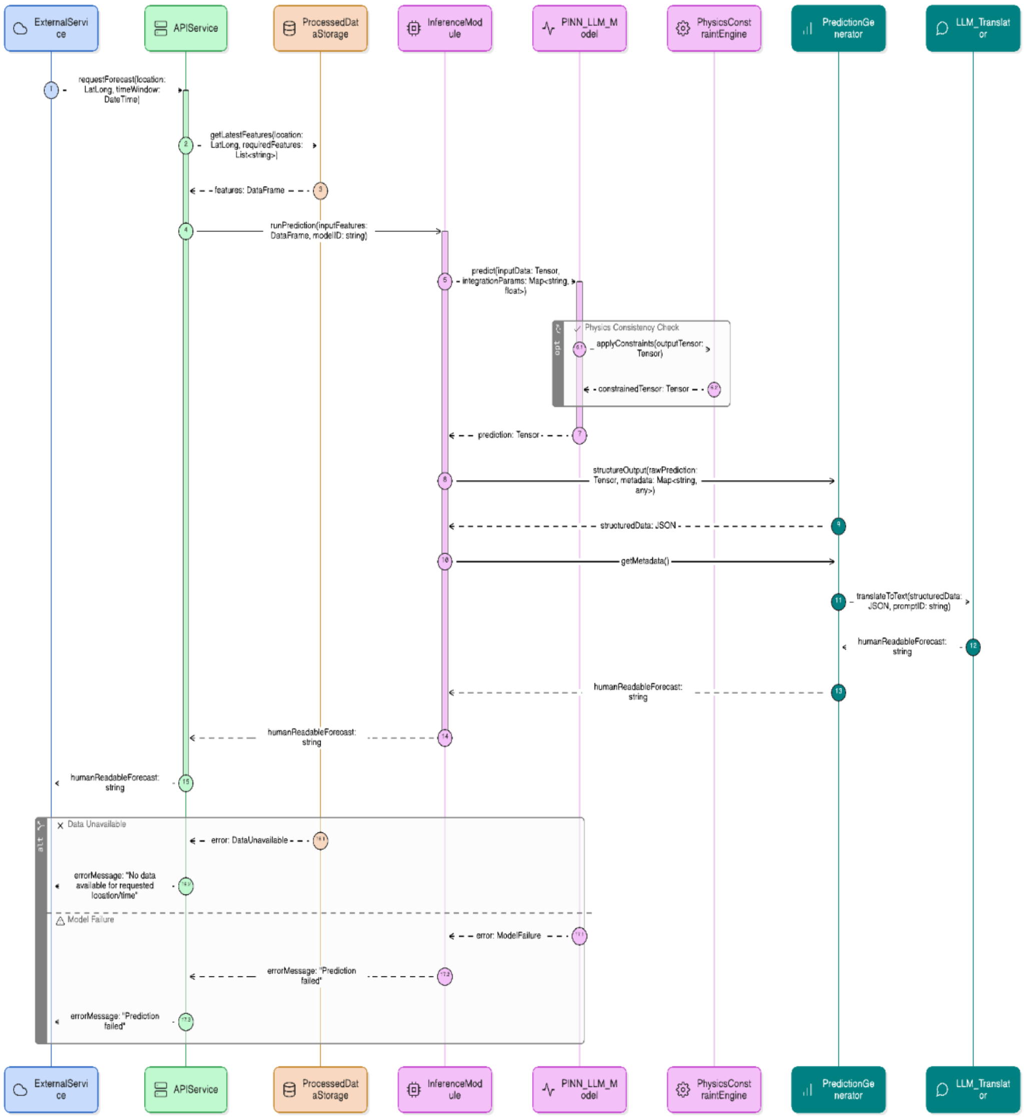
5.2. ER Diagram



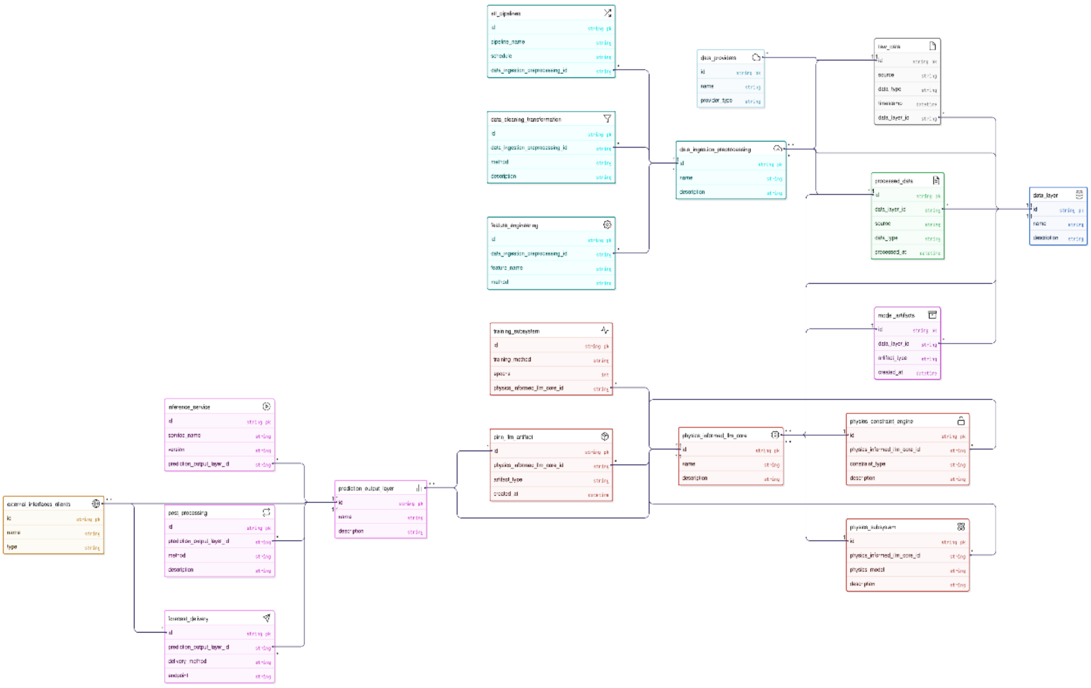
5.3. State Transition Diagram



5.4. Sequence Diagram

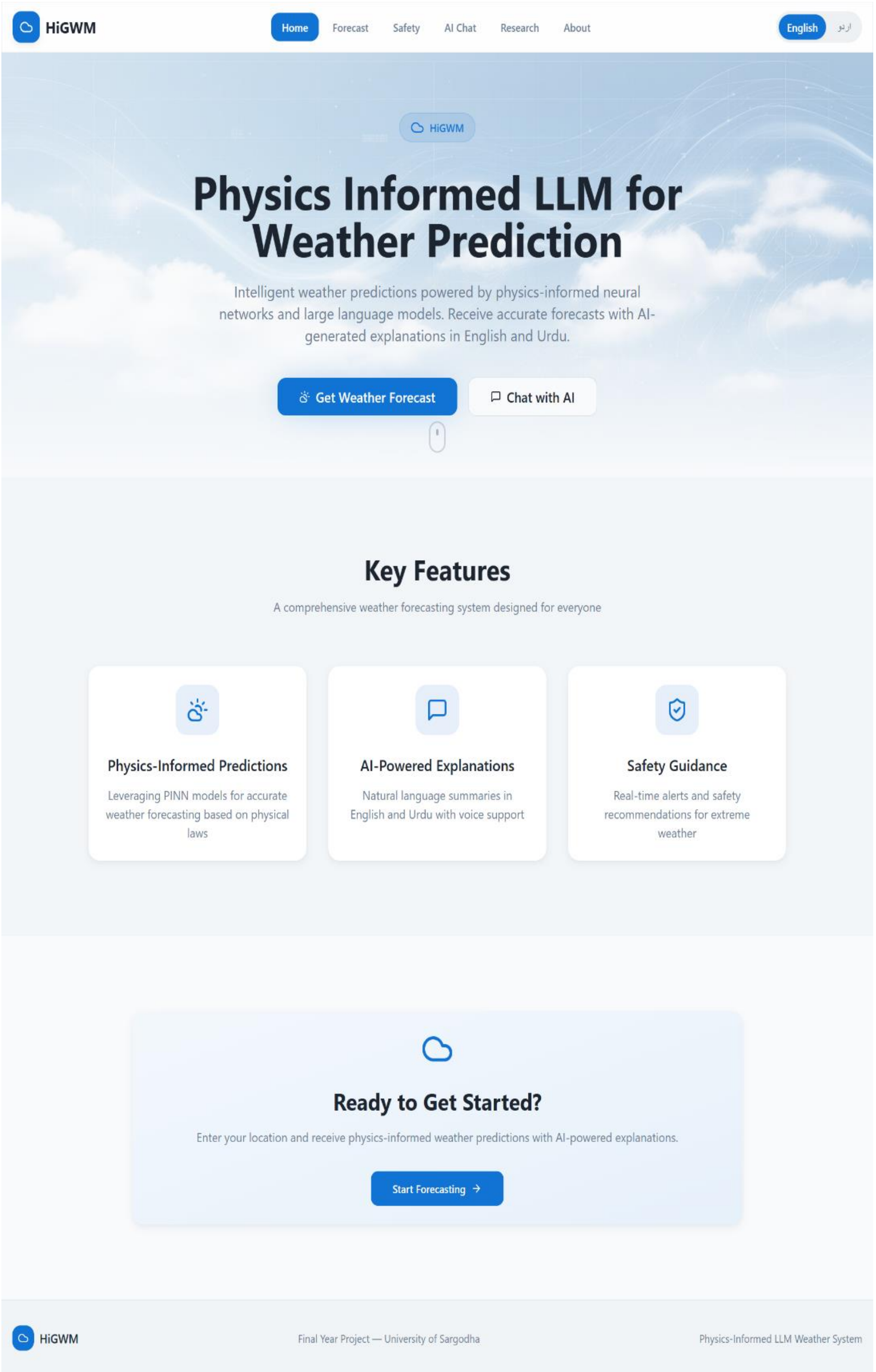


5.5. Package Diagram



5.6. GUI

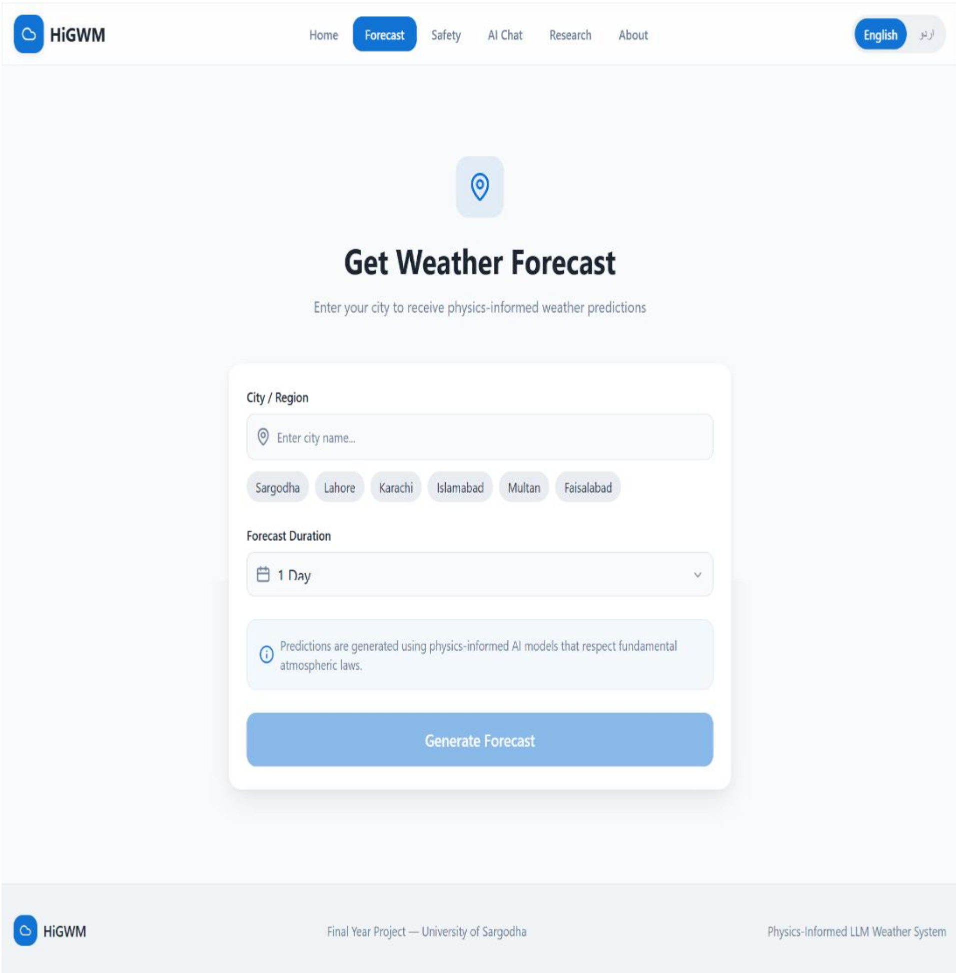
5.6.1. Home Page



Entry point of Application is the Home Page. It delivers a clear and brief overview of the **Physics-Informed LLM Based Weather Prediction System** and introduces users to the system's capabilities and purpose. The interface presents the project title, a short description, and quick navigation options to access weather forecasting and AI chat features. The Home Page contains the project title, a short description about the project and a navigation bar to access other options.

The page is designed with a clean and minimal layout to ensure ease of use for both technical and non-technical users. It also supports bilingual content (English and Urdu), allowing users to switch languages seamlessly.

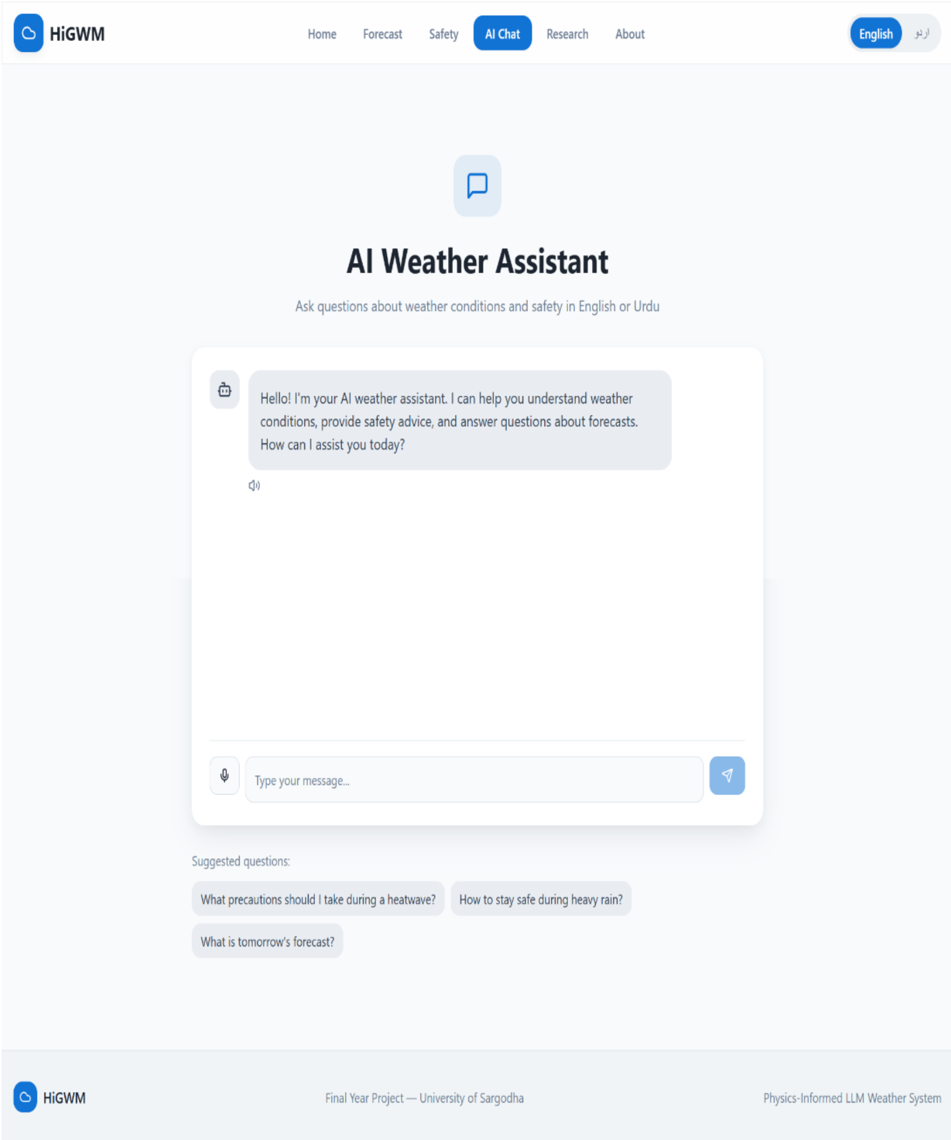
5.6.2. Forecast Page



The Forecast Page allows users to request and view weather predictions for a selected location. Users can input any city or region of Pakistan and duration of forecast. As an output, system will display numerical weather parameters like temperature, humidity, pressure and wind speed in an organized form called dashboard format.

The other feature this project provides is summary of weather forecast that explain the forecast in natural language and in our case natural language can be Urdu and English.

5.6.3. LLM Page



The AI Chat Page allows user to perform an interactive communication with the LLM. Different questions can be asked related to weather conditions and safety measures, specially in extreme weather conditions.AI will answer the question in human like style which will be easy to understand.

This page supports bilingual interaction (English and Urdu) and is developed to help user in understanding weather risks and recommend precautions.

6. References

Ref. No.	Document Title	Date	Source
[1]	Proposal_G38-SS1	2025	https://github.com/Abubakr4t/FYP-/blob/main/Proposal_G38-SS1.pdf
[2]	SRS-1.3	2025	https://github.com/Abubakr4t/FYP-/tree/main/SRS

7. Appendix

Workflow: Weather data → PINN forecast → LLM explanation → User output.